Additive coding and smaller coefficients for Diophantine consistency equations

An explicit synthesis for Con(ZFC)

Research report prepared for ProveIt

AI-assisted mathematical synthesis, construction, and independent audit

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Abstract

We construct a completely specified ordinary polynomial in ten unrestricted integer unknowns whose unsolvability is equivalent, provably in ZFC, to Con(ZFC) for the repository's proof predicate. Its degree is exactly 107 286 763 576, compared with $126496 \cdot 5^{31} + 27064$ for the ten-integer construction in the pinned repository. The degree decreases by a factor of approximately $5.49 \cdot 10^{15}$. The arithmetic circuit grows from 1,442 to 2,204 binary operations. This is a degree improvement at a fixed number of unknowns.

The new step replaces exponentially spaced positions 5^j by a finite additive B_4 code. Quartic coefficient recovery needs only distinct sums of at most four positions. A translated Bose–Chowla set supplies 32 positions, including the homogenizing zero, with maximum 942302. We prove the exact coefficient identity, bounds, mask decoding, and the complete source interface required by Sun's ten-integer reduction.

Independently, a native compiler refinement encodes every edge label by a nonzero word, making separate occupancy bits redundant. Together with an audited support layout, this reduces the native base from the previous smallest bound $2^{1012115}$ to 2^{683195}. The four fixed coefficient files occupy 342,269 raw bytes. We bind this tuple to the existing P67 equation and 59-operation system. We also deliver a fully composed ten-integer variant, regenerating all source coefficients and bounds from the new tuple. It retains the same degree and circuit size and reduces its five raw fixed payloads from 1,066,835 to 685,209 bytes. A separate comparison instance preserves the original tuple and isolates the additive-code improvement.

The archive includes the report source and PDF, complete circuits, actual fixed coefficients, reproducible generators, finite certificates, independent reviews, and provenance. No global minimality, new universal variable-count theorem, consistency proof, or new end-to-end Lean/Rocq certificate is claimed.

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1 The results and what they improve

All natural numbers include zero. A positive variable is strictly positive. An unrestricted integer variable may have any sign. These domains are part of each theorem.

Let $\mathrm{Ref}_{\mathrm{ZFC}}$ denote the existence of a finite refutation in the proof calculus used by the pinned ProveIt development. Thus Con(ZFC) abbreviates $\neg\,\mathrm{Ref}_{\mathrm{ZFC}}$ for that calculus.

Theorem 1.1 (The comparison ten-integer equation)**.** *The comparison circuit* `ten_integer/generated/b4_signed10.json` *defines a polynomial*

$$F_{\mathrm{B4}}(f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}[f,g,h,k,l,w,x,y,z,m]$$

such that

$$\mathrm{ZFC} \;\vdash\; (\exists \boldsymbol{u} \in \mathbb{Z}^{10} : F_{\mathrm{B4}}(\boldsymbol{u}) = 0) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}\,.$$

Consequently,

$$\boxed{\mathrm{ZFC} \;\vdash\; (\nexists \boldsymbol{u} \in \mathbb{Z}^{10} : F_{\mathrm{B4}}(\boldsymbol{u}) = 0) \longleftrightarrow \mathrm{Con(ZFC)}.}$$

The exact degree is 107286763576*. The supplied evaluator uses 485 additions, 25 subtractions, and 1,694 multiplications: 2,204 binary arithmetic operations in total.*

The degree decrease is

$$\frac{126496 \cdot 5^{31} + 27064}{107286763576} = \frac{589042901992797851562527064}{107286763576} \approx 5.4903595034 \cdot 10^{15}.$$

There are 762 more arithmetic operations than in the previous ten-integer circuit, an increase of about 52.8%. The new method improves degree dramatically while retaining the same ten unrestricted unknowns and a compact, finite expression.

The recommended complete instance is `ten_integer_composed/generated/b4_signed10.json`. Corollary 5.1 proves the same equivalence, degree, and operation count for this variant, with all fixed coefficients and the source-base bound recomputed from the smaller tuple below. The comparison circuit remains included to separate the two mathematical changes.

The second result concerns a different measure:

Theorem 1.2 (A smaller complete native tuple)**.** *The nonzero-edge compiler in* `coefficient_backend/` *supplies an admissible fixed blank-K215 tuple* $(B-1, K, M_C, M_F^{\mathrm{src}})$ *with*

$$B = 2^{683195}, \qquad \max\{\textit{port bit lengths}\} = 688546.$$

Its four materialized unsigned integer files contain 342,269 raw bytes. Substitution of this entire tuple in the inherited P67 equation and retained 59-operation system preserves their equivalence to blank K215 halting, and hence their unsolvability is equivalent in ZFC to Con(ZFC)*.*

The source base exponent decreases by 32.50% compared with package 36's 1012115; raw bytes decrease by 32.45% from 506,664. These percentages compare complete tuples. The proof does not splice the smallest individual ports from different constructions.

1.1 There is no single smallest measure

Variable count, degree, number of arithmetic operations, coefficient size, and proof complexity are distinct. The present work supplies two new points in that comparison. It does not dominate the old 67-operation equation or the nine-natural-variable constructions on every measure.

Construction	Domain	Degree	Operations
Pinned package 31	10 integers	$126496 \cdot 5^{31} + 27064$	1442
This report, main circuit	10 integers	107286763576	2204
Pinned P67; new tuple here	15 positive	155	67
Pinned quartic, package 31	31 positive	4	125
Pinned package 34	12 positive	1022	124
Pinned package 37	9 positive	$92324 \cdot 5^{32} + 296$	1500

The pinned values are the audited collection's stated values, supported by their source formulas. The new circuits and coefficient metrics are recomputed in this delivery. Nine-natural or nine-positive results are not ten-integer results with a different notation.

2 Repository analysis and the inherited boundary

The source revision is

`22e75dd6b87b1b199bb08c778d7cf075ae5aae08`.

Its collection contains wave 5, packages 31–37, with a subsequent audit correcting several comparisons. Earlier waves remain recoverable from history. In particular, the relevant baseline already has ten unrestricted integers via Sun's 2026 preprint [5]; improving to fourteen or twelve unrestricted integers would not advance that frontier.

2.1 What was combined

Package 31 provides the explicit ten-integer outer construction, its quintic divisibility merger, the coprime positivity merger, and economical separating weights. It also supplies a complete fixed coefficient tuple and the 31-variable quartic. These are retained in the main result.

The new additive code replaces the source polynomial's exponentially spaced digit positions. It is not an optimization of a few gates in the final norm. It changes the exponent geometry at the stage that determines the enormous degree.

The coefficient result combines the Wang943 native compiler, the actual-rank control method refined in package 36, a new nonzero-word encoding of edge labels, and an exact support layout. It improves explicit coefficient size while preserving the simple P67 and 59-operation arithmetic skeletons.

The inspections of packages 34 and 37 also identify two relevant boundaries: their shifted twelve-positive-variable norms are different polynomials whose positive completeness requires a proof, and their small signed Jones forms rely on additional cofinality arguments. Neither of those delicate transformations is needed for the principal ten-integer result here.

2.2 The fixed parent

The positive variables, in their authoritative order, are

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

The four fixed ports are (β, K_0, M_C, M_F), with M_F already the shifted mask. Set

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C = q - F - \alpha.$$

Appendix A gives all 22 quadratic residuals r_j. For the nonnegative coding domain define

$$P_0(n_1, \ldots, n_{31}) = \sum_{j=1}^{22} r_j(n_1 + 1, \ldots, n_{31} + 1)^2. \tag{1}$$

This is a quartic. Its constant term is strictly positive because at the all-one parent assignment the residual $E - X(a - E)$ is 1. Every zero therefore has at least one nonzero coordinate.

The comparison tuple is package 31's complete 121-by-8801 tuple, whose native base is $2^{1064921}$. Its four materialized ports have bit lengths

$$1064921, \quad 1070004, \quad 1064921, \quad 1064922.$$

The compressed files and `ten_integer/data/INSTANCE.json` fix them exactly. Historical metadata in a byte-preserved parent file does not override this active binding. Section 5 binds the same parent equations to the smaller tuple and recomputes all following coefficient bounds.

For

$$P_0(\boldsymbol{n}) = \sum_{|\alpha| \le 4} p_\alpha \boldsymbol{n}^\alpha, \qquad c_\alpha = (4 - |\alpha|)! \prod_{i=1}^{31} \alpha_i! \, p_\alpha,$$

the exact reconstruction gives 402 nonzero records and proves

$$|c_\alpha| + 1 < 2^{4269855}.$$

The source radix multiplier is the actual fixed integer

$$\beta_S = 2^{4269882},$$

which is also materialized and hashed. Since $48 \cdot 33^4 = 56924208 < 2^{27}$, it satisfies

$$\beta_S > 48 \cdot 33^4 \max_\alpha |p_\alpha|.$$

All coefficient records are finite integer expressions in the four fixed ports. There is no unspecified recursively enumerable index or existential coefficient.

2.3 The endpoint being inherited

The pinned development provides

$$\exists \boldsymbol{n} \in \mathbb{N}^{31} : P_0(\boldsymbol{n}) = 0 \quad \Longleftrightarrow \quad \text{K215 halts on the blank tape} \quad \Longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}\,. \tag{2}$$

The formal machine theorem is `KEndpoint215.blank_halts_iff_zfc_refutation`, whose statement uses `BlankHaltsmachine215` and `BProvZFCax_s[]Form.fBot`. The native compiler and quartic adapters remain ordinary mathematical developments. Their particular proof files and the coefficient adapter are cited and retained with provenance.

We use the mathematical equivalences represented by those developments. We do not infer a ZFC theorem from Lean's soundness, and we do not claim a new build of the formal machine endpoint.

3 Additive coding of the quartic source

The main new construction replaces the positional exponents in the ten-integer equation of package 31 by an additive code. The source quartic, its complete coefficient tuple, and the outer Lucas and relation-combining formulas are retained. In particular, the ten-integer theorem remains Sun's theorem [5]; the result below is an explicit improvement of its source specialization [2]. Bose–Chowla codes themselves are classical [6].

Theorem 3.1 (The explicit ten-integer instance)**.** *The circuit* `ten_integer/generated/b4_signed10.json` *defines a polynomial*

$$F_{B_4} \in \mathbb{Z}[f, g, h, k, l, w, x, y, z, m]$$

of exact total degree

$$\boxed{\deg F_{B_4} = 107\,286\,763\,576.}$$

Its supplied division-free circuit has 2 204 *binary additions, subtractions, and multiplications. With the fixed parent quartic and compiler tuple specified in this report, ordinary mathematics formalizable in ZFC proves*

$$\exists (f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}^{10}\ (F_{B_4} = 0) \quad \Longleftrightarrow \quad \neg\,\mathrm{Con}(\mathrm{ZFC}).$$

Consequently absence of integer zeros is equivalent, in ZFC, to Con(ZFC).

The proof imports the same K215 calibration, native compiler, positive quartic reduction, and Sun bridge as the pinned construction. It gives a new proof of the entire source-coding interface. Exact finite checks supplement that proof; they are not an end-to-end formal verification of the imported results. Sun's ten-integer bridge is cited in its 2026 preprint version [5]. No consistency proof, globally minimal equation, or new general ten-variable theorem is claimed.

3.1 The fixed source and its coefficient convention

Let $r_1, \dots, r_{22}$ be the fixed positive-parent residuals specified in this report, in their literal order, and put

$$P_0(n_1, \dots, n_{31}) = \sum_{j=1}^{22} r_j(n_1 + 1, \dots, n_{31} + 1)^2.$$

The inherited endpoint states

$$\exists \mathbf{n} \in \mathbb{N}^{31}\ P_0(\mathbf{n}) = 0 \quad \Longleftrightarrow \quad \text{ZFC has a finite syntactic refutation.} \tag{B4.1}$$

Here $\mathbb{N}$ includes zero. The positive-to-natural shift is part of the definition and cannot be omitted.

The four fixed coefficient ports are the complete Wang943 tuple of package 31, with port order (β, K_0, M_C, M_F) and bit lengths

$$(1064921,\ 1070004,\ 1064921,\ 1064922).$$

The high mask M_F is already shifted. None of these ports is existentially quantified, and no port is substituted from another tuple. The exact parent and compressed integer payloads are included in `ten_integer/data/`.

The source has degree four and $P_0(\mathbf{0}) > 0$. For the latter fact, the fifth residual $E - X(a - E)$ equals 1 at the all-one parent assignment. Write

$$P_0(\mathbf{n}) = \sum_{|\alpha| \le 4} p_\alpha \mathbf{n}^\alpha, \qquad c_\alpha = (4 - |\alpha|)! \prod_{i=1}^{31} \alpha_i!\, p_\alpha, \tag{3}$$

$$H = \max_\alpha |p_\alpha|, \qquad \beta_S = 2^{4269882}. \tag{4}$$

The included coefficient certificate proves

$$\beta_S > 2 \cdot 24 \cdot 33^4 H. \tag{B4.2}$$

Indeed it bounds every weighted coefficient by $|c_\alpha| + 1 < 2^{4269855}$, and $2 \cdot 24 \cdot 33^4 = 56924208 < 2^{27}$. Since the integer factorial multiplier in (3) is positive, $|p_\alpha| \le |c_\alpha|$. Thus (B4.2) follows without requiring decimal expansion of the enormous coefficients.

The source has 402 nonzero weighted coefficient records. Each is an explicit small polynomial in the four fixed ports. The verifier reconstructs every record from the 22 shifted residual squares. The leading zero retained in each serialized exponent vector is a historical coefficient-interface convention: it is not an input digit, an unknown, or an extra source witness.

3.2 What the exponent code must distinguish

Definition 3.2. A list $0 = w_0 < w_1 < \cdots < w_{31}$ is a quartic additive code if all sums of four list elements, allowing repetition and disregarding order, are distinct.

Padding a multi-index with $4 - |\alpha|$ copies of w_0 shows that this condition is equivalent to injectivity of

$$\alpha \longmapsto \alpha \cdot \mathbf{w} := \sum_{i=1}^{31} \alpha_i w_i \qquad (|\alpha| \le 4). \tag{B4.3}$$

The positional weights 5^i satisfy this property. They are not needed for either the witness-digit mask or coefficient extraction. An additive code gives much smaller exponents.

The code used here is

$$\begin{aligned} &0,\ 42609,\ 49910,\ 91724,\ 155840,\ 213052,\ 290241,\ 337708,\\ &340502,\ 351998,\ 388348,\ 417182,\ 480443,\ 513128,\ 534148,\\ &563475,\ 579595,\ 636647,\ 639228,\ 692126,\ 725139,\ 732225,\\ &748536,\ 754497,\ 817940,\ 820267,\ 829985,\ 858576,\ 870324,\\ &\qquad\qquad 879730,\ 907673,\ 942302. \end{aligned}$$

The ith positive entry is assigned to the ith coordinate of the literal parent ordering. Throughout this section set

$$T = 942302, \qquad L_c = 4T = 3769208, \qquad M = 2^{20} - 1 = 1048575. \tag{B4.4}$$

Lemma 3.3 (Finite code certificate)**.** *The displayed list is a quartic additive code. In fact, its four-term multiset sums are distinct modulo M.*

Proof. There are $\binom{35}{4} = 52360$ relevant multisets. The included short checker enumerates all of them and verifies distinct residues modulo M. This provides a direct finite certificate independent of the construction history.

For a mathematical construction, form the binary quotient by $X^{20} + X^3 + 1$ and let θ be the residue of X. Exact modular computations give $\theta^M = 1$ and

$$\theta^{M/p} \neq 1 \qquad (p = 3, 5, 11, 31, 41),$$

the complete set of prime divisors of M. Hence θ has order M. Its M distinct powers and zero exhaust the 2^{20} residues. Every nonzero residue is therefore a unit, proving that the quotient is a field, rather than assuming irreducibility in this step.

Let $\mathbb{F}_{32}$ be its 32-element subfield and choose the 32 logarithms a_c satisfying

$$\theta^{a_c} = \theta + c \qquad (c \in \mathbb{F}_{32}).$$

The element θ has degree four over $\mathbb{F}_{32}$. If two sums of four logarithms agree modulo M, the corresponding products of four factors $\theta + c$ agree. Subtracting their monic degree-four polynomials leaves a polynomial of degree at most three over $\mathbb{F}_{32}$. It must vanish identically, so unique factorization identifies the two multisets of factors. This is the Bose–Chowla argument [6].

Translate the logarithms by -535448 modulo M and take their representatives in $[0, M)$. The sorted list is the displayed code. Translation preserves equality or inequality of four-term sums modulo M. The chosen origin follows the largest cyclic gap of the original logarithms and reduces the maximum representative to T. The generator reproduces each finite-field calculation, all logarithms, the translation, and the exhaustive multiset certificate. □

3.3 The coefficient identity

Define formal polynomials

$$C_{\mathbf{n}}(t) = \left(1 + \sum_{i=1}^{31} n_i t^{w_i}\right)^4, \qquad D_c(t) = \sum_{|\alpha| \le 4} c_\alpha t^{L_c - \alpha\cdot\mathbf{w}}. \tag{5}$$

Every exponent of D_c is nonnegative, since $\alpha \cdot \mathbf{w} \le 4T = L_c$.

Lemma 3.4 (Quartic coefficient recovery)**.** *For every* $\mathbf{n}$*,*

$$[t^{L_c}] C_{\mathbf{n}}(t) D_c(t) = 24 P_0(\mathbf{n}). \tag{B4.5}$$

If $0 \le n_i < b$ *and* q_j *denotes any coefficient of* $C_{\mathbf{n}}(t)D_c(t)$*, then*

$$|q_j| \le 24H \left(1 + \sum_i n_i\right)^4 \le 24H\big(1 + 31(b-1)\big)^4. \tag{B4.6}$$

The product has degree at most $8T$*.*

Proof. The multinomial term indexed by γ in $C_{\mathbf{n}}$ has coefficient

$$\frac{4!}{(4 - |\gamma|)! \prod_i \gamma_i!}\, \mathbf{n}^\gamma$$

and exponent $\gamma \cdot \mathbf{w}$. Its product with the α-term of D_c reaches exponent L_c exactly when $\gamma \cdot \mathbf{w} = \alpha \cdot \mathbf{w}$. Lemma 3.3 forces $\gamma = \alpha$, and the factorials cancel. Summing proves (B4.5).

Distinct indices in D_c have distinct exponents and coefficients of magnitude at most $24H$. The coefficients of $C_{\mathbf{n}}$ are nonnegative and their sum is $C_{\mathbf{n}}(1)$. Bounding each convolution coefficient by $24HC_{\mathbf{n}}(1)$ proves (B4.6). Finally, $\deg C_{\mathbf{n}} \le 4T$ and $\deg D_c \le L_c = 4T$. □

3.4 The general mechanism behind the improvement

The coefficient step is not special to the particular quartic support. It explains why the change can be much larger than a local circuit simplification.

Proposition 3.5 (Polynomial span at fixed degree)**.** *Let* $d \ge 2$ *and let* $P \in \mathbb{Z}[x_1, \dots, x_n]$ *have total degree at most* d*. For every prime power* $q \ge n + 1$ *there are fixed integer weights*

$$0 = w_0 < w_1 < \cdots < w_n \le T \le q^d - 2$$

such that, on putting

$$c_\alpha = (d - |\alpha|)! \prod_i \alpha_i!\, p_\alpha, \qquad D_P(t) = \sum_{|\alpha| \le d} c_\alpha t^{dT - \alpha\cdot\mathbf{w}},$$

one has the exact identity

$$[t^{dT}] \left(1 + \sum_{i=1}^{n} x_i t^{w_i}\right)^d D_P(t) = d!\, P(\mathbf{x}).$$

In particular, choosing the least power of two $q \ge n + 1$ *gives* $T < [2(n+1)]^d$*.*

Proof. The Bose–Chowla construction in $\mathbb{F}_{q^d}$ gives a set of q residue classes modulo $q^d - 1$ with distinct unordered d-sums. Indeed a primitive element has degree d over $\mathbb{F}_q$; equality of products of d factors $\theta + c$ yields an identically zero polynomial of degree at most $d - 1$, exactly as in Lemma 3.3. Translate one class to zero, retain it and any n other classes, and take their representatives in $[0, q^d - 2]$. Padding with zero gives the needed injectivity for all multi-indices of total degree at most d. The multinomial proof of Lemma 3.4, with $d!$ in place of 24, proves the displayed identity. Finally $q < 2(n+1)$ for the stated choice of q. □

Thus at fixed d the required exponent span is polynomial in the number of source variables, whereas the standard positional choice has span $(d+1)^n$. This statement concerns the coefficient-coding stage. A full Diophantine reduction still requires its bounds, masks, and both implications to be checked. The following proof performs that work for the delivered quartic instance. The finite-field theorem is classical; no claim of literature priority for this general use of additive codes is made.

3.5 The new two-witness source interface

For integer variables f, g, define the following expressions in the displayed order:

$$b = 1 + 3f, \qquad \mathcal{B} = \beta_S b^4, \qquad \ell = \sum_{i=1}^{31} \mathcal{B}^{w_i}, \tag{6}$$

$$D_c = D_c(\mathcal{B}), \qquad N_0 = \mathcal{B}^{T+1}, \qquad N_1 = 4\mathcal{B}^{8T+1}, \tag{7}$$

$$N = N_0 N_1, \qquad \mathscr{C} = b\mathcal{B}^T, \qquad M_c = N_0 - 1 - (b-1)\ell, \tag{8}$$

$$T_c = M_c + (\mathcal{B} - 2)\mathcal{B}^{L_c} N_0, \tag{9}$$

$$J_c = 2(1+g)^4 D_c + \sum_{j=0}^{8T} \mathcal{B}^{j+1}, \tag{10}$$

$$S_c = g + J_c N_0, \qquad R_c = (S_c + T_c + 1)N + T_c + 1, \tag{11}$$

$$X = (N-1)R_c, \qquad Y = N^2. \tag{12}$$

All exponents and sum limits are fixed integers. These expressions are ordinary polynomials in f, g with fixed integer coefficients.

Theorem 3.6 (Complete source interface)**.** *The following implications hold.*

1. *If P_0 has a natural zero, then for every positive bound Z there exist $f \geq Z$ and $b \leq g < \mathscr{C}$ such that b is a square power of two and*

$$Y \mid \binom{2X}{X}. \tag{B4.7}$$

2. *If $f \neq 0$, $0 \leq g < 2\mathscr{C}$, b is a square power of two, and (B4.7) holds, then P_0 has a natural zero.*

3. *Already under $f \neq 0$, b square, and $0 \leq g < 2\mathscr{C}$, without a power-of-two assumption on b,*

$$3 \mid X, \qquad X > 3b, \qquad Y > \max\{b, 256\}. \tag{B4.8}$$

Moreover $\mathscr{C} = 2^{4269882T} b^{4T+1}$, with an even positive exponent of two, so the bound has exactly the form required by Sun's source interface.

Proof. We first establish signs and bounds before assuming a source zero or binomial divisibility. Since $b = 1 + 3f$ is a square and $f \neq 0$, one has $f > 0$ and $b \geq 4$. The coefficient bound (B4.2) implies

$$\frac{\mathcal{B}}{2} > 24H\big(1 + 31(b-1)\big)^4, \tag{13}$$

$$\mathcal{B} > 2 + 2 \cdot 24H(32b)^4. \tag{14}$$

For the second inequality, replacing 33 by 32 in the bound for β_S leaves more than two units of margin after multiplication by b^4, because $H \geq 1$ and $b \geq 4$.

The constant coefficient $P_0(\mathbf{0})$ is positive. Thus $D_c > 0$: its leading term is $24P_0(\mathbf{0})\mathcal{B}^{L_c}$ and the absolute sum of all lower terms is at most

$$24H \sum_{j=0}^{L_c-1} \mathcal{B}^j < \mathcal{B}^{L_c}.$$

We also have $D_c \le 24H \sum_{j=0}^{L_c} \mathcal{B}^j$. The base-$\mathcal{B}$ digits of M_c are $\mathcal{B}-b$ at positions w_i and $\mathcal{B}-1$ at the other positions from 0 through T. Therefore

$$0 \le M_c < N_0, \qquad 0 < (\mathcal{B}-2)\mathcal{B}^{L_c} < N_1, \qquad 0 \le T_c < N. \tag{B4.9}$$

For $0 \le g < 2\mathscr{C}$, the inequalities $2b \le \mathcal{B}$ and $g \in \mathbb{Z}$ give $g < N_0$ and $1 + g \le 2b\mathcal{B}^T \le 32b\mathcal{B}^T$. Using (14) and the bound for D_c,

$$\begin{aligned} 0 < J_c &< \frac{(\mathcal{B}-2)+\mathcal{B}}{\mathcal{B}-1}\mathcal{B}^{8T+1} \\ &= 2\mathcal{B}^{8T+1} < N_1. \end{aligned}$$

Consequently $0 \le S_c < N$. These are unconditional bounds on the stated source range.

Because β_S is an even power of two and $b \equiv 1 \pmod 3$, we have $\mathcal{B} \equiv 1 \pmod 3$ and $N = 4\mathcal{B}^{9T+2} \equiv 1 \pmod 3$. Hence $3 \mid X$. Also $R_c \ge N+1$, $N > \mathcal{B} \ge b$, and $N \ge 4$, so $X = (N-1)R_c > 3b$ and $Y = N^2 > \max\{b, 256\}$. This proves (B4.8).

Now assume additionally that b is a power of two. Then $\mathcal{B}, N_0, N_1, N$ are powers of two. For nonnegative integers a, d, write $\tau_2(a,d) = 0$ when their binary addition has no carries, equivalently when their bitwise AND is zero. The mask description gives

$$0 \le g < 2\mathscr{C}, \quad \tau_2(g, M_c) = 0 \iff g = \sum_{i=1}^{31} n_i \mathcal{B}^{w_i} \quad \text{for some } 0 \le n_i < b. \tag{B4.10}$$

Indeed the digit $\mathcal{B}-b$ has ones exactly in the high binary positions of one $\mathcal{B}$-digit, whereas $\mathcal{B}-1$ has ones in every position. The only free bits of g are therefore the low $\log_2 b$ bits at the distinct slots w_i. The bound $g < N_0$ excludes all higher slots. Conversely every displayed code has the required bit pattern and is less than $\mathscr{C}$. No superincreasing property of the weights is used.

For decoded g, write

$$(1+g)^4 D_c = \sum_{j=0}^{8T} q_j \mathcal{B}^j$$

using the formal coefficients of Lemma 3.4, and put

$$J' = \sum_{j=0}^{8T} \left(q_j + \frac{\mathcal{B}}{2}\right) \mathcal{B}^j.$$

Equations (B4.6) and (13) show that every displayed digit of J' is an integer strictly between 0 and $\mathcal{B}$. In particular this is its actual base-$\mathcal{B}$ expansion, even when some q_j are negative. Since $J_c = 2J'$,

$$\tau_2(J_c, (\mathcal{B}-2)\mathcal{B}^{L_c}) = 0 \iff \tau_2\left(J', \left(\frac{\mathcal{B}}{2}-1\right)\mathcal{B}^{L_c}\right) = 0.$$

The mask on the right occupies precisely the low bits of the digit at L_c. A digit in $[1, \mathcal{B})$ avoids it exactly when it equals $\mathcal{B}/2$. By (B4.5), this is equivalent to $q_{L_c} = 24P_0(\mathbf{n}) = 0$.

The two block decompositions, together with $S_c, T_c < N$, give

$$\tau_2(S_c, T_c) = 0 \iff \tau_2(g, M_c) = 0 \ \wedge \ \tau_2(J_c, (\mathcal{B}-2)\mathcal{B}^{L_c}) = 0. \tag{B4.11}$$

Sun's binary two-block lemma [4, Lemma 2.2] states, for $0 \le S_c, T_c < N$ with N a power of two and the exact R_c, X above, that

$$\tau_2(S_c, T_c) = 0 \quad \Longleftrightarrow \quad N^2 \mid \binom{2X}{X}. \tag{B4.12}$$

Its hypotheses have all been established before invoking it. Combining (B4.10)–(B4.12) proves soundness.

For completeness, suppose $P_0(\mathbf{n}) = 0$. Choose s so large that $b = 2^{2s} > \max_i n_i$ and $(b-1)/3 \ge Z$. Since $4^s \equiv 1 \pmod 3$, $f = (b-1)/3$ is an integer. The source zero is not the all-zero vector, so $g = \sum_i n_i \mathcal{B}^{w_i} \ge \mathcal{B} \ge b$. The usual geometric-sum estimate gives $g < b\mathcal{B}^T = \mathscr{C}$. The mask and coefficient identities yield (B4.7). Finally the asserted monomial formula for $\mathscr{C}$ follows directly from $\mathcal{B} = \beta_S b^4$. □

3.6 The complete outer polynomial

We now give the full outer formula, so that the artifact is specified by fixed constants, finite sums, and ordinary polynomial operations. Only the ten lowercase coordinates listed in Theorem 3.1 are unknowns. Every other symbol is a circuit register.

Starting with the new $b, X, Y, \mathscr{C}$ above, set

$$\begin{aligned} L_s &= lY, & U &= 2L_sX, \\ V &= 4gwY, & W_s &= bw, \qquad (15) \\ A &= U(V+1), & B_s &= 2X+1, \\ C &= B_s + (A-2)h, & P_n &= U^2V, \qquad (16) \\ K &= X + 1 + k(P_n - 2), & S_s &= 2A - 5, \qquad (17) \\ T_s &= 3W_sC - 2(W_s^2 - 1), & R_2 &= (P_n^2 - 4)K^2 + 4, \qquad (18) \end{aligned}$$

$$\Delta = 8\mathscr{C}^3 g K^2 - g^2\{32(C - KL_s)^2\mathscr{C}^3 + g^2K^2\}, \tag{19}$$

$$O = f^2 l^2 \Delta, \qquad \xi = zx. \tag{20}$$

The modified Lucas certificate is

$$\begin{aligned} D_s &= (A^2-4)C^2 + 4, & E &= C^2 D_s \xi, \qquad (21) \\ F_s &= 4(A^2-4)E^2 + 1, & G_* &= 2(A+2)(A-2)^2E^2 - 1, \qquad (22) \\ \Omega &= 16E, & H_* &= C + (B_s - C)F_s + \Omega F_s y, \qquad (23) \\ I_* &= (G_*^2 - 1)H_*^2 + 1. \qquad (24) \end{aligned}$$

These are the same bridge and Lucas formulas as in package 31 [2], using Sun's modified certificate and interpolation theorem [5].

For the divisibility and nonvanishing merger, define

$$\begin{aligned} r &= T_s + S_s y, & \Sigma &= \xi S_s^5, \qquad (25) \\ \Theta &= (r^3 - rS_s^2 - S_s^3)(r^2 + 23S_s^2). \qquad (26) \end{aligned}$$

The three radicands and their separating weights are

$$\begin{aligned} r_0 &= b\{1 + (3O-4)b^2z^2\}, & r_1 &= R_2, & r_2 &= D_sF_sI_*, \qquad (27) \\ \omega_0 &= 1, & \omega_1 &= r_0, & \omega_2 &= r_0(P_n^2 + K^2 + 3). \qquad (28) \end{aligned}$$

Finally put

$$Z_c = \Sigma m - \Theta, \qquad A_i = \Sigma^2 \omega_i^2 r_i \quad (0 \le i \le 2), \tag{29}$$

$$V_n = Z_c^2 + A_2 - A_0 - A_1, \tag{30}$$

$$W_n = V_n^2 + 4Z_c^2 A_2 - 4A_0 A_1, \tag{31}$$

$$\boxed{F_{B_4}} \boxed{= W_n^2 - 16 V_n^2 Z_c^2 A_2.} \tag{32}$$

The requested equation is $F_{B_4} = 0$. No radical in the proof below is part of its definition. There are no divisions or variable exponents in (6)–(32).

3.7 Why the finalizer retains both implications

For clarity we record the inherited finalizer arguments, with their attribution to package 31 [2]. They ensure that the new source interface is being composed with the precise outer hypotheses.

Lemma 3.7 (The quintic merger)**.** *The polynomial $\pi(v) = (v^3 - v - 1)(v^2 + 23)$ has a root modulo every positive integer and no rational root. If $S_s \neq 0$ and $\xi \neq 0$, then $\Sigma \mid \Theta$ implies $S_s \mid T_s$. Conversely, when $S_s \mid T_s$, a residue class of y modulo $|\xi|$ makes $\Sigma \mid \Theta$. For every integer assignment with $S_s \neq 0$, $\Theta \neq 0$.*

Proof. The cubic has discriminant -23. At $p = 23$ it has the simple root 3. At an odd prime $p \neq 23$, either it has a simple root or it is irreducible. In the latter case Frobenius is an even three-cycle, so the discriminant is a square modulo p and $v^2 + 23$ has a simple root. Hensel lifting handles all odd prime powers. Since $-23 \equiv 1 \pmod 8$, the quadratic has a root in the 2-adic integers as well. The Chinese remainder theorem gives a root modulo every positive integer.

The rational-root theorem excludes rational roots of the cubic, whose only possible integer roots are ± 1; the quadratic is positive on $\mathbb{Q}$. Consequently $\Theta = S_s^5 \pi((T_s + S_s y)/S_s)$ cannot vanish.

If $S_s \mid T_s$, write $T_s = S_s q$ and choose $y \equiv c - q \pmod{|\xi|}$ for a root c of π modulo $|\xi|$. Then $\Theta = S_s^5 \pi(q + y)$ is divisible by Σ. Conversely suppose that a prime satisfies $a = v_p(S_s) > d = v_p(T_s)$. Then $v_p(r) = d$; the leading term of the homogeneous cubic factor has uniquely smallest valuation $3d$, and that of the quadratic factor has uniquely smallest valuation $2d$. Thus $v_p(\Theta) = 5d < 5a$, contradicting $S_s^5 \mid \Theta$. Therefore $S_s \mid T_s$. □

Lemma 3.8 (Positivity and separating weights)**.** *For the displayed polynomial expressions, if $z \neq 0$ then $|r_0| > 1$ and*

$$\sqrt{|r_1|} \le P_n^2 + K^2 + 2. \tag{B4.13}$$

Any rational signed sum $\pm\sqrt{r_0} \pm \omega_1\sqrt{r_1} \pm \omega_2\sqrt{r_2}$ forces every r_i to be an integer square. Moreover, r_0 square and $z \neq 0$ imply that b is a positive square and $O > 0$.

Proof. The congruence $b \equiv 1 \pmod 3$ excludes $b = 0, -1$. If $r_0 = \pm 1$, its factor b must be 1, hence $f = O = 0$ and $r_0 = 1 - 4z^2$, a contradiction. If $r_0 = 0$, its bracket vanishes, forcing $|b| = |z| = 1$ and $3O - 4 = -1$; again $b = 1$ gives $O = 0$, a contradiction. This proves $|r_0| > 1$.

For (B4.13), use $|r_1| \le (P_n^2 + 4)K^2 + 4$. Subtracting this bound from $(P_n^2 + K^2 + 2)^2$ leaves $P_n^4 + K^4 + P_n^2 K^2 + 4P_n^2 \ge 0$. It follows that

$$|\omega_1| > \sqrt{|r_0|}, \qquad |\omega_2| > \sqrt{|r_0|} + |\omega_1|\sqrt{|r_1|}. \tag{B4.14}$$

In the multiquadratic splitting field, any automorphism changing a nonzero term of a rational signed sum has a largest changed index. The magnitude of that term is at least the magnitude of its weight, since its nonzero radicand is an integer. By (B4.14), all earlier terms together are too small to cancel its change. This contradiction shows that every root term is rational. A rational

square root of an integer is integral. The argument also covers negative candidate radicands by complex absolute values, and zero radicands require no recovery.

Finally the two factors of r_0 are coprime, since the bracket is 1 modulo b. If their product is a nonzero square, each factor is a signed square with the same sign. A negative square cannot be 1 modulo 3, so b is a positive square and its bracket is a square. If $O \le 0$ and $bz \neq 0$, that bracket would be at most $1 - 4 = -3$. Hence $O > 0$. □

Proof of Theorem 3.1: equivalence. The polynomial identity underlying the finalizer is

$$F_{B_4} = \prod_{\varepsilon \in \{-1,1\}^3} \left(Z_c + \sum_{i=0}^{2} \varepsilon_i \sqrt{A_i} \right). \tag{B4.15}$$

It follows by successively multiplying conjugate factors, or by expanding the 35 terms of the three-radicand norm. The delivered checker proves this exact formal identity independently.

Suppose $F_{B_4} = 0$. The integer $S_s = 2A - 5$ is odd and nonzero, so Lemma 3.7 gives $\Theta \neq 0$. If $\Sigma = 0$, then every $A_i = 0$ and (32) equals $\Theta^8 \neq 0$. Thus $\Sigma \neq 0$, and $\xi = zx \neq 0$. Divide a vanishing factor of (B4.15) by Σ. Lemma 3.8 makes every r_i a square; the resulting signed sum of roots is an integer, so $\Sigma \mid \Theta$. Lemma 3.7 gives $S_s \mid T_s$, and Lemma 3.8 gives b square and $O > 0$. These, $\xi \neq 0$, R_2 square, and $D_s F_s I_*$ square are exactly the hypotheses of Sun's reverse bridge [5, Proposition 6.2]. It yields the source range, the power-of-two condition, and binomial divisibility required by Theorem 3.6. Hence P_0 has a natural zero, and (B4.1) yields a finite ZFC refutation.

Conversely suppose P_0 has a natural zero. Theorem 3.6 supplies arbitrarily large f; choose $f \ge 2$. Sun's positive bridge [5, Proposition 6.1] supplies the other initial witnesses with b square, R_2 square, $S_s \mid T_s$, $C = u_{B_s}(A)$, and $O > 0$. As $O = f^2 l^2 \Delta$ is a positive integer, $O \ge 4$. The positive nonsquare integer $d = 3O - 4 \equiv 2 \pmod 3$ has a Pell solution with its second coordinate a positive multiple of b: take a positive Pell unit to a power equal to the order of its class modulo b. Write that second coordinate as bz with $z > 0$. Then r_0 is a square.

Sun's full residue-class interpolation theorem [5, Theorem 4.1] supplies a positive ξ divisible by this z, and allows y in the residue class required by Lemma 3.7, while making $D_s F_s I_*$ a square. Set $x = \xi / z$. Thus all r_i are squares, $\Sigma \neq 0$, and $\Sigma \mid \Theta$. Choose the integer m so that a selected signed factor of (B4.15) vanishes. This gives $F_{B_4} = 0$.

All new steps have been proved above. The substantial imported machine/compiler and Lucas results are explicitly identified; their ordinary arithmetic proofs, not the soundness of any proof assistant, supply the asserted ZFC-provability. □

3.8 Circuit identity, degree, and reproducibility

The sparse coefficient evaluator uses the division-free identity

$$\mathcal{B}^{4T - \alpha \cdot \mathbf{w}} = (\mathcal{B}^T)^{4 - |\alpha|} \prod_{i=1}^{31} (\mathcal{B}^{T - w_i})^{\alpha_i}. \tag{B4.16}$$

All exponents are nonnegative. A multivariate Horner scheme shares the powers and coefficient recipes. Equation (B4.16) is valid for arbitrary integer assignments, including $\mathcal{B} = 0$; no division by $\mathcal{B}$ is concealed. The geometric sum in (10) uses binary identities for fixed powers and fixed-length sums.

The main circuit contains 485 additions, 25 subtractions, and 1694 multiplications, for 2204 operations. There are 79 fixed coefficient leaves after reuse and literal simplification. The general source-degree bounds in this instance are

$$\deg X = 108T + 24, \qquad \deg Y = 72T + 16, \qquad \deg \mathscr{C} = 4T + 1. \tag{B4.17}$$

Propagating degree bounds through the circuit yields

$$\deg F_{B_4} \le 113856T + 27064 = 107286763576. \tag{B4.18}$$

To certify attainment rather than just an upper bound, substitute one indeterminate for all ten unknowns. With the actual fixed coefficient payloads, the coefficient at the bound in (B4.18) is 16 modulo 101. It is also 599 modulo 1009 and 622356 modulo 1000003. A single nonzero such coefficient proves attainment, so the stated total degree is exact. The checker performs exact leading-coefficient arithmetic modulo the primes; this is not inference from random numerical agreement.

For comparison, package 31's positional construction has exact degree

$$126496 \cdot 5^{31} + 27064 = 589042901992797851562527064$$

and 1442 operations. The new degree is smaller by a factor of approximately 5.49×10^{15}, at the cost of 762 additional arithmetic operations. Thus the degree improvement is substantive, while the operation-count tradeoff remains explicit.

The self-contained commands are

```
python3b4_code.py
python3build_b4_sun10.py
python3verify_b4_sun10.py
```

from `ten_integer/`. They use only the Python standard library. In addition to the degree proof, they regenerate the finite field and all 52360 code sums, reconstruct all 402 weighted coefficient records, check the fixed coefficient payloads and bounds, and expand the norm identity exactly. A separate scalar transcription evaluates the literal 402-term coefficient sum and all named registers modulo three primes, independently of the Horner circuit. Those modular transcription checks are regression evidence; the preceding lemmas supply the universal equivalence.

The archive also includes four finalizer variants using more generic weights or the literal Sun four-square merger. They share the same source adapter and fixed tuple. A clean isolated rebuild reproduces every generated JSON file byte for byte. Source pins, coefficient bindings, licenses, and inherited attribution notices accompany the construction.

4 Nonzero label codes and smaller fixed coefficients

The variable-compression construction and the coefficient construction address different measures. This section gives a new coefficient tuple and binds two complete, moderate-degree equations to it. It does not silently substitute these coefficients into a low-variable form whose index constants were computed from another parent polynomial.

4.1 The new tuple and its measured improvement

The baseline is package 36 of the reviewed repository commit. It uses 943 Wang tiles, a radix $\rho = 2^{115}$, cell length 8801, and $B = 2^{1012115}$. The new construction uses the identical tile language, but has

$$b = 107, \qquad L = 6385, \qquad d = bL = 683195, \qquad \rho = 2^{107}, \qquad B = \rho^L = 2^{683195}. \tag{33}$$

All four resulting coefficients are materialized, with their exact unsigned little-endian payloads and hashes included in the archive. The comparison is:

Measure	Package 36	New tuple
Base exponent $\log_2 B$	1,012,115	683,195
Largest coefficient, bits	1,016,946	688,546
Four raw payloads, bytes	506,664	342,269

These decrease by 32.50%, 32.29%, and 32.45%, respectively. There are two mathematical changes: an occupancy bit on each edge word is replaced by nonzeroness of that whole word; and exact support isolation permits a much shorter synchronization layout.

4.2 Occupancy requires a nonzero word, not an extra bit

Nonzero-code lemma. *Suppose a local selector/copy clause permits either an empty cell, whose edge words are all zero, or one selected tile with its four encoded edge words. Suppose the horizontal and vertical label encodings are injective and never zero, and corresponding words agree at neighboring cells. A Start tile in a cyclic word whose spatial shift is one forces every cell to be occupied and decodes the original tile matching rules.*

Indeed, a neighbor of an occupied cell has a matching nonzero word and therefore cannot be empty. Iteration of spatial step one visits the whole cyclic word. Once occupancy is established, injectivity decodes the original labels. Conversely, any tile matching supplies the encoded words and satisfies all bit comparisons. This argument does not require a bit position shared by every nonzero word.

The inherited horizontal colors are $0, \dots, 249$, and the vertical colors are $0, \dots, 438$. Encode every label x as $x + 1$. Horizontal words now use eight bits and vertical words nine. This is optimal *within the fixed-length, separate-edge encoding interface with zero reserved for emptiness*: m labels require $2^\ell - 1 \geq m$. The bounds are therefore $\ell_H \geq 8$ and $\ell_V \geq 9$, attained here. This is not a lower bound on general Diophantine constructions.

4.3 The entire finite recipe

Keep the ordered 943 Wang tiles, with Start at selector 0 and Forbidden at selector 1. The language and literal 430-action K215 table are unchanged. The separately supplied tile list has SHA-256

```
66b55f1d11fca9fefc93b12fcfc8b4cabb75
    b08e9fd03311f9e894a1c5e5e81f.
```

Use the following slot data:

Slot s	Offset o_s	Width ℓ_s	Field p_s
West	1023	8	0
North	1013	9	8
East	1003	8	17
South	983	9	25

The native support E is the disjoint union of $\{0, \dots, 942\}$, the 34 copy positions $o_s + j$, the dummies $1063, 1064$, and the anchors $1022, 1042, 2107$. Hence $|E| = 982$. Set $w_j = 2^{10+2j}$ for $0 \leq j < 37$. For tile i, with label $x_{i,s}$ in slot s, define

$$c_i = 1 + \sum_{s=0}^{3} \sum_{j=0}^{\ell_s - 1} \mathrm{bit}_j(x_{i,s} + 1) w_{p_s + j} + w_{34} + w_{35} + w_{36}.$$

At the copy positions set $c_{o_s+j} = w_{p_s+j}$; at the three anchors use w_{34}, w_{35}, w_{36}, in the displayed order; and set both dummy coefficients to zero. Define the following fixed integers:

$$\mu = 1022 + \sum_{j=0}^{36} w_j,$$

$$Z = (2^{105} - 1) \sum_{e=6376}^{6383} \rho^e + (2^{30} - 1)\rho^{6384},$$

$$D_C = \rho^{30} + \rho^{70} + \rho^{20} + \rho^{1085} + \sum_{e \in E} c_e(\rho^{3202-e} + \rho^{4268-e}),$$

$$D_R = \rho^{50},$$

$$M_C = B - 1 - \sum_{\substack{e \in E \\ e \neq 1}} \rho^e - 2\rho^{1063},$$

$$M_{F,\text{native}} = 4 + \mu(\rho^{3202} + \rho^{4268}) + \sum_{j=0}^{8} \rho^{1013+j} + \sum_{j=0}^{7} \rho^{1073+j} + \rho^{1042} + \rho^{2107} + Z.$$

The four actual ports are

$$\boxed{\beta = B - 1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F = M_{F,\text{native}} + B - 1.} \tag{34}$$

In particular M_F already includes its shift. All sums and powers in this recipe are fixed finite arithmetic; none is an existential variable or a computation that waits for a refutation.

4.4 Why the new finite interface is sound and complete

Let n be the selector count in one native cell. Before inspecting any payload field, $0 \le n \le 943$, and the lowest ten bits of the clause sum are precisely n. Disjointness with 1022 forces $n = 0$ or $n = 1$. Only then does each radix-four payload field become a sum in $\{0, 1, 2\}$. Testing its low bit forces equality of the selected tile bit and its copy, and forces each anchor to equal n. Thus payload carry-freeness is not assumed before selector extraction.

The complete comparison identities are:

Target	Current source via D_C	Spatial source	Temporal source
$1013 + j$, $0 \le j < 9$	$983 + j$	none	$1013 + j$
$1073 + j$, $0 \le j < 8$	$1003 + j$	$1023 + j$	none
1042	1022	none	1042
2107	1022	none	2107

At each clause band $T = 3202, 4268$, the coefficient of ρ^{T-e} in D_C is exactly c_e for every $e \in E$, and $e+50 \neq T$. The generator and independently written auditor check all $19 \cdot 982 \cdot 3 + 2 \cdot 982 \cdot 2 = 59{,}902$ scalar identities. Their linearity covers every native assignment, including the allowed dummy digits. No additional bare or reflected-clause contribution is omitted.

Write $S = E \setminus \{2107\}$. Its maximum is 1064. The difference 1066 is absent from $E - E$, since pairs in S have smaller span and $2107 - 1066 = 1041 \notin E$. Because $L > 2 \max E$,

$$((3202 - E) \bmod L) \cap ((4268 - E) \bmod L) = \varnothing.$$

At least one clause band is clean for every binary rotation. Decode the distinguished Start cell first: its selector is already one, so its source anchor at 1022 is one and both tested temporal anchors must be one. This step does not presuppose global occupancy.

The source's parity argument excludes nonzero binary remainders: for remainders $1, \ldots, b-2$ every rotated native digit is even, and for $b - 1$ only the single upper-dummy spill can be odd.

Two tested anchors cannot both be odd in either case. The remaining whole-radix alignment is unique, since the difference $2107 - 1042 = 1065$ occurs exactly once in $E - E$. Thus

$$((1042 - E) \bmod L) \cap ((2107 - E) \bmod L) = \{0\}.$$

After alignment, the parity comparisons give equality of corresponding bits. The nonzero-code lemma then propagates occupancy and recovers the unchanged marked Wang tiling. Conversely every intended tiling supplies these words and anchors, so the change loses no tableau.

The inherited conservative mass bound is

$$\mathcal{M} = (982 + 2)\left(2\sum_{e \in E} c_e + 6\right).$$

Its bit length is 104, while $4\mathcal{M}+8$ has bit length 106; therefore $4\mathcal{M}+8 < 2^{107}$. Also $2\mu+4 < 2^{107}$ and $2(2^{105} - 1) + 4 < 2^{107}$. These are the inherited uniform carry and strict-slack bounds. The maximum unshifted current exponent is $4268 + 2107 = 6375$, and the spatial maximum is $50 + 2107 = 2157$. All padding positions $6376, \ldots, 6384$ are consequently zero on every intended aligned field, including safe dummy variations.

The unpadded high-mask population is $2 \cdot 46 + 9 + 8 + 2 + 1 = 112$. Padding contributes $8 \cdot 105 + 30 = 870$, giving

$$\mathrm{pc}(M_{F,\mathrm{native}}) = 982, \qquad \mathrm{pc}(M_C) = d - 982.$$

Moreover $M_C \equiv 2 \pmod 4$, $M_{F,\mathrm{native}} \equiv 4 \pmod 8$, and both native masks lie strictly between zero and $B - 1$. The origin test is unchanged because every exponent in D_C, D_R is positive.

4.5 Controlling the actual Pell rank

Changing d requires a new actual-rank controller. We have

$$d = 5 \cdot 107 \cdot 1277, \qquad \lambda = 135257 \text{ prime}, \qquad k = \lambda - 1 = 135256.$$

The orders of B modulo $5, 107, 1277, \lambda$ are respectively $4, 106, 1276, 67628$. Hence $\mathrm{ord}_d(B) = 67628 \mid k$, while

$$B^k \bmod \lambda^2 = 13998558473 \neq 1, \qquad v_\lambda(B^k - 1) = 1.$$

For $H_0 = 1 + BD_C + B^2(D_R + 1)$, the exact unit certificate is

p	$B \bmod p$	$(B^2 - 1) \bmod p$	$H_0 \bmod p$
5	3	3	2
107	88	39	58
1277	942	1125	956
135257	55321	88158	66933

Choose independently padded width w_0 and height h_0 to be sufficiently large powers of λ, put $N = w_0 h_0 = \lambda^n$, $q = B^N$, and require $N > d\lambda^2$ and $h_0 > \lambda$. Then $q \equiv B^{w_0} \equiv B \pmod{d\lambda}$. Flipping a low dummy at cell j changes the actual packed rank by

$$\delta R = -\Gamma B^j, \qquad \Gamma = \rho^{1063}(q^2 - 1)[1 + q(D_C + BD_R + B^{w_0})].$$

The table proves $\gcd(\Gamma, dN) = 1$.

For completeness, put $T_0 = N/\lambda$ and $C_0 = d\lambda$. Every residue modulo $dN = C_0 T_0$ is a Boolean subset sum of

$$\{B^{2+kj} : 0 \leq j < T_0\} \cup \{B^3\}.$$

Indeed the first T_0 powers of B^k are exactly the residues $1 + C_0 z$. Normalize a target by B^{-2} and subtract ϵB, $\epsilon \in \{0, 1\}$, to make it nonzero modulo λ. Its least residue a modulo C_0 satisfies $1 \le a < C_0 < T_0$ and $\gcd(a, T_0) = 1$. Writing the target as $a + C_0 u$, solve

$$a z_0 + \frac{a(a-1)}{2} \equiv u \pmod{T_0}.$$

The a distinct orbit entries with additive coordinates $z_0, z_0 + 1, \ldots, z_0 + a - 1$ have the desired sum. Include B^3 if $\epsilon = 1$. The greatest selected position is at most $N - N/\lambda - \lambda + 3$, and $w_0 < N/\lambda$; hence no required spatial or temporal translate wraps or touches the origin.

Starting from rank R_0, choose the subset for $(R_0 - dw_0)\Gamma^{-1}$ modulo dN. It gives $R \equiv dw_0 \pmod{dN}$, exactly the temporal rotation required by $X = 2^R$. The strict positive slack survives every safe dummy choice. The inherited Pell converse supplies the remaining witnesses. Prime powers are a completeness choice only; soundness does not impose them on arbitrary solutions.

Both d and N are odd, and the canonical word is 1 modulo 4. The balanced populations and low congruences therefore preserve the inherited identities $\mathrm{pc}(R) = 3dN + 2$ and $R \equiv 3 \pmod 4$. In particular X/q remains a positive square, and the inherited odd Pell multipliers and odd half-binomial scale remain available. This also verifies the completeness promises of the compatible signed arithmetic adapters.

4.6 Two concrete equations bound to the new tuple

To make the coefficient result a complete mathematical instance, the archive supplies the old arithmetic graphs with all four ports rebound to (34). The 15 positive variables for the first are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma.$$

The following are polynomial abbreviations, not extra unknowns:

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, \\ X &= wq, & Y &= sq^3, \\ E_0 &= XY, & k_0 &= \eta + \zeta, \\ c &= k_0 Y + \eta, & a &= E_0 + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H, \\ D &= X + ac + \gamma H, & G &= (q-1)(q-F) + \alpha. \end{aligned}$$

Also set

$$\begin{aligned} R &= G(q^2 - 1) + (M_C + qM_F)J, \\ V &= c(Tf - 1) - Rf^2, \qquad K_a = (i\Delta c^2)^2. \end{aligned}$$

Set

$$\begin{aligned} N_0 &= \tau^2 - (XY^2)(XY^2 + 1)k_0^2, \\ N_1 &= D^2 - \Delta c^2, \\ N_a &= K_a(V^2 - y^2) + y^2, \\ N_k &= k_0 - hE_0 - R, \\ N_t &= (K + w)C + (q - F) - z(q - 1), \\ N_s &= \Delta f^2 - K_a. \end{aligned}$$

Then the fixed equation is

$$\boxed{P_{67}^{\mathrm{nz}} = N_0 N_1 N_a N_k N_t N_s - \Delta = 0.} \tag{35}$$

Its delivered graph has 67 binary arithmetic operations, comprising 39 multiplications and 28 additions/subtractions. Its exact degree is 155. The crucial cancellation is the polynomial identity

$$N_1 = 2ac(X + \gamma H) + (X + \gamma H)^2 - Hc^2,$$

which gives degree 18. The six factor degrees are $22, 18, 60, 7, 2, 46$. For example, the degree-155 monomial

$$J^{92} h \gamma i^4 \eta^{13} w^{13} s^{26} T^2 f^2 z$$

has coefficient $8\beta^{92} \neq 0$. Thus the degree remains exact after the new fixed tuple is bound.

For the second presentation replace T by two positive unknowns o, j, so that there are sixteen unknowns in total, and retain the same outer abbreviations. Put

$$L_0 = E_0 k_0 Y, \quad V_0 = of - c, \quad U_0 = jc - R, \quad I = ic^2, \quad A_0 = \Delta(f^2 - 1).$$

The seven equations are

$$\begin{aligned} \tau^2 - L_0(L_0 + k_0) &= 1, & D^2 - \Delta c^2 &= 1, \\ A_0(V_0^2 - y^2) + y^2 &= 1, & k_0 - hE_0 - R &= 1, \\ (K + w)C &= F + z(q - 1), & I^2 &= A_0, \\ U_0 &= V_0. \end{aligned} \tag{36}$$

Their residual degrees are $22, 18, 24, 7, 2, 22, 6$. The graph evaluates the comparisons using 59 arithmetic operations, comprising 33 multiplications and 26 additions/subtractions; comparisons are free under the repository convention. These are inherited arithmetic graphs, not new operation records.

Bound-instance theorem. *ZFC proves that each of* (35) *and* (36) *has no solution in its stated positive-integer domain if and only if* Con(ZFC).

The inherited Pell/arithmetic proofs equate solutions of these graphs with the blank marked native compiler problem, provided its complete four-port interface holds. The preceding argument establishes that interface for the new tuple, including soundness, actual-rank tuning, strict positive witnesses and canonical signed promises. The unchanged Wang-language theorem equates a marked tiling with a finite blank K215 halting run, and the K215 calibration equates such a run with a finite ZFC refutation. Composition and negation prove the displayed theorem. This is a proof of an equivalence, with the inherited mathematical dependencies made explicit; it is neither a decision of consistency nor a new end-to-end kernel check.

4.7 Reproducible evidence and limits

The directory `coefficient_backend/` contains the full recipe, literal raw and gzip coefficients, machine actions, independently pinned tiles, preserved notices, inherited proof sources, and the new proof. Its builder and auditor require only the Python standard library. The independent review found no gap in the nonzero-code bootstrap or the new synchronization support.

The concrete circuit verifier proves that every arithmetic graph, witness list and comparison is unchanged from its pinned source, and that every coefficient port is fixed to the declared payload hash. It reproduces exact degree bounds using the main-norm cancellation, and proves degree attainment by a nonzero leading coefficient on a specified line modulo 1000000007. This latter computation is an exact lower-degree certificate, not probabilistic identity testing. All 67 or 59 gates and all four ports are live.

The archived `PROOF.md` gives the complete adapter derivation, including the Boolean subset-sum controller. Finite checks verify its explicit constants and support identities; they do not by themselves prove the unbounded Pell, tiling or K215 calibration theorems. The comparison ten-integer compression retains its original parent tuple and fixed indices. Section 5 completes the rebinding as a distinct, fully regenerated ten-integer instance.

5 A complete combined instance

The two improvements can be composed without changing the number of unknowns or the arithmetic graph. The separate directory `ten_integer_composed/` performs this composition explicitly: it retains the literal quartic parent and the B_4 code, substitutes all four ports of the nonzero683195 backend together, recomputes every source coefficient bound, and materializes a new β_S. The original `ten_integer/` instance remains available for the isolated degree comparison.

Corollary 5.1 (The combined ten-integer equation)**.** *The fixed circuit* `ten_integer_composed/generated/b4_signed10.json`*, with its local* `data/INSTANCE.json` *binding, defines a polynomial* $F_{B_4}^{\mathrm{nz}} \in \mathbb{Z}[f,g,h,k,l,w,x,y,z,m]$ *satisfying*

$$\mathrm{ZFC} \;\vdash\; \Big[(\exists \mathbf{u} \in \mathbb{Z}^{10} : F_{B_4}^{\mathrm{nz}}(\mathbf{u}) = 0) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}\Big].$$

Thus its lack of integer zeros is equivalent in ZFC to Con(ZFC)*. Its exact degree is* 107286763576 *and its circuit has* 2204 *binary operations. Its native base is* 2^{683195} *and its source-margin constant is*

$$\boxed{\beta_S^{\mathrm{nz}} = 2^{2743514}.}$$

The four native ports together with this margin constant occupy 685209 *raw bytes, compared with* 1066835 *for the comparison ten-integer instance.*

This corollary uses the same imported K215, ordinary compiler/Pell, and Sun theorems as the main construction. The new work is the fully specified composition and its coefficient and degree audit.

5.1 The positive parent is compatible with the new tuple

The literal parent polynomial is unchanged, including its old positive transport coordinate z_{old}. To verify compatibility, use the sixteen positive variables of the new retained59 system and write its transport coordinate as z_{new}. Define

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha,\\
X &= wq, & Y &= sq^3,\\
E &= XY, & k &= \eta + \zeta,\\
c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H,\\
D &= X + ac + \gamma H.
\end{aligned}$$

The packed rank is

$$R = ((q-1)(q-F) + \alpha)(q^2-1) + (M_C + qM_F)J.$$

Also put $V = of - c$, $I = ic^2$, $A_0 = \Delta(f^2-1)$ and $L_0 = EkY$. Denote the retained residuals by

$$\begin{aligned}
u_1 &= \tau^2 - L_0(L_0+k) - 1, & u_2 &= D^2 - \Delta c^2 - 1,\\
u_3 &= A_0(V^2 - y^2) + y^2 - 1, & u_4 &= k - hE - R - 1,\\
u_5 &= (K_0 + w)C - F - z_{\mathrm{new}}(q-1), & u_6 &= I^2 - A_0,\\
u_7 &= jc - R - V.
\end{aligned}$$

These are the seven equations already bound to the complete new tuple.

Extend a positive retained solution by

$$\begin{aligned} z_{\mathrm{old}} &= z_{\mathrm{new}} + 1, & N &= q^3, & L &= q^2 - 1, \\ P &= E(c-\eta) + k, & C_2 &= c^2, & F_2 &= f^2, \\ B_{\mathrm{aux}} &= I^2, & g &= y^2 - V^2, & & \end{aligned}$$

and by the displayed expressions for the other parent auxiliaries. The source coordinate y is unchanged. These parent coordinates are distinct from the later ten-integer coordinates with the same letter names.

The two sign obligations. All added auxiliaries are immediately positive except V and g. The inherited ordinary-kernel bootstrap gives

$$c > A\Delta^2, \qquad A = a + 2, \qquad \Delta = A^2 - 1.$$

This is equation (3) of Section 4 in the byte-preserved source `source_proofs/ORDINARY_BLANK_PROOF.md`. It occurs before auxiliary rank recovery: the first-norm index is at least 24, the main index is larger, and Pell growth gives the displayed bound. In particular $c > 2$ and $c^2 > 4\Delta$. Here the ordinary strong equation is

$$I^2 = \Delta(f^2 - 1), \qquad I = ic^2.$$

It therefore gives

$$f^2 = 1 + \frac{i^2 c^4}{\Delta} > 4c^2.$$

Consequently $V = of - c > c > 1$, since o is a positive integer. The auxiliary equation now yields

$$I^2 V^2 - (I^2 - 1)y^2 = 1, \qquad g = y^2 - V^2 = \frac{V^2 - 1}{I^2 - 1} > 0.$$

These inequalities hold for every positive retained solution.

There is no identification of different normalized multipliers: the parent uses the ordinary i. In a canonical P67 witness, $i_{\mathrm{P67}} = \psi_A(m)/c^2$, whereas the corresponding ordinary parent coordinate is $i = \Delta i_{\mathrm{P67}}$. The direct retained59 binding avoids conflating the two strong equations.

Exact residual identities. Under the displayed extension, all 22 parent residuals vanish identically except positions $8, 12, 13, 14, 15, 21, 22$. In order, these become

$$u_1, \quad u_2, \quad u_4 - u_7, \quad u_7, \quad u_5, \quad u_6, \quad u_3 + (V^2 - y^2)u_6.$$

The only packing simplification is the polynomial identity

$$(q - F)(N - q) - C(q^2 - 1) = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) \qquad (N = q^3).$$

The transport identity is

$$(K_0 + w)C + q - F - z_{\mathrm{old}}(q-1) - 1 = (K_0 + w)C - F - z_{\mathrm{new}}(q-1).$$

Thus a positive retained solution extends to a positive literal parent solution.

Conversely, a positive parent zero has $q > 1$. Its old transport equation implies

$$(K_0 + w)C = F + (z_{\mathrm{old}} - 1)(q-1) > 0,$$

hence $C > 0$ and $F < q$. Restore the deleted packing register

$$G = q(q - F) - C = (q-1)C + q\alpha > 0.$$

The packing identity in the inherited quartic31 proof restores the complete positive source. Its ordinary kernel gives $X = wq = 2^R$ and $R > 3q$, so $w > q > F$. Therefore

$$(z_{\text{old}} - 1)(q - 1) = (K_0 + w)C - F > 0.$$

It follows that $z_{\text{new}} = z_{\text{old}} - 1$ is strictly positive. The same identities recover the seven retained equations. This establishes the positive zero-set equivalence with all auxiliaries and transport conventions accounted for.

The nonzero backend has already established the complete arithmetic interface needed by this inherited kernel. The positive parent therefore has a zero precisely when there is a finite refutation. Replacing its coordinates by $n_i + 1$ gives the required source quartic on $\mathbb{N}^{31}$. Its value at zero remains positive because the fifth residual is 1 at the all-one parent assignment.

5.2 Recomputing the source margin

The new fixed port bit lengths are

$$(b_1, b_2, b_3, b_4) = (683195, 688546, 683195, 683196).$$

The symbolic coefficient recipes are unchanged. For a weighted coefficient written $c_\alpha = \sum_e a_{\alpha,e} \prod_j A_j^{e_j}$, use the explicit triangle bound

$$U_\alpha = \sum_e |a_{\alpha,e}|\, 2^{\sum_j e_j b_j}.$$

Since $A_j < 2^{b_j}$, $|c_\alpha| \le U_\alpha$. All 402 individual bounds are recomputed from these recipes and the actual port payloads. Their largest certified exponent gives

$$|c_\alpha| + 1 < 2^{2743487}.$$

As before $|p_\alpha| \le |c_\alpha|$. Because $2 \cdot 24 \cdot 33^4 = 56924208 < 2^{27}$, the new even power $\beta_S^{\text{nz}} = 2^{2743514}$ satisfies the full strict source-margin hypothesis. The new β_S is materialized; its raw length is 342940 bytes. Adding the four native payloads gives $342269 + 342940 = 685209$ bytes, a 35.77% reduction against the five payloads of the comparison ten-integer construction.

The code weights, source interface, and outer relation-combining formula now apply verbatim with the new values. All required properties have been re-established: a fixed quartic in 31 natural variables, positive constant term, the exact factorial coefficient convention, the B_4 code, and the strict even-power margin. No new existential parameter is introduced.

5.3 Exact degree and reproducible composition

The delivered arithmetic DAG is identical to the comparison DAG: 485 additions, 25 subtractions and 1694 multiplications, with 10 unknown leaves and 79 fixed coefficient leaves. The new fixed values cannot raise its degree bound. To exclude cancellation lowering it, the verifier computes the diagonal leading coefficient with the actual new payloads:

prime	101	1009	1000003
leading coefficient	36	63	258712

Every entry is nonzero. Thus

$$\deg F_{B_4}^{\text{nz}} = 113856T + 27064 = 107286763576.$$

The script `regenerate_binding.py` reads the sibling complete backend and base parent, copies the four coherent payloads, recomputes all coefficient bounds and β_S, and rewrites all active binding metadata. It is followed by `b4_code.py`, `build_b4_sun10.py` and `verify_b4_sun10.py`. The independent review scripts separately reconstruct the coefficients, check the byte binding, and audit the formula and degree without importing these builders. The source proofs and their Git blob identities are included. The finite certificates supplement the universal transfer proof; they do not replace the inherited arithmetic or logical theorems.

6 Verification and proof status

A zero-equivalence theorem is an assertion about all integer assignments. Finite tests alone cannot establish it. The universal new source argument and coefficient adapter are therefore proved as ordinary mathematics, and the finite checks certify the specific data and arithmetic claims used by those proofs.

For the main ten-integer equation, the executable verifier:

1. reconstructs the shifted sum of the 22 parent residual squares and all 402 weighted coefficients exactly;
2. verifies the literal compressed ports, source hashes, and all coefficient bounds;
3. checks all 52,360 four-multiset sums of the code;
4. expands the norm identities exactly in a formal polynomial algebra, with 35 terms for three radicands and 495 terms for four;
5. compares each emitted circuit with the literal formulas modulo three primes, including named intermediate quantities;
6. verifies the ten variable leaves, binary gate ledger, fixed-leaf bindings, and an exact degree-attainment certificate.

For the original-tuple comparison circuit, substituting a single indeterminate for all unknowns gives a coefficient at degree 107286763576 equal to 16 modulo 101. Since every gate also satisfies the advertised degree upper bound, this nonzero residue proves exact degree. The residues 599 modulo 1009 and 622356 modulo 1000003 provide additional checks. One nonzero residue already suffices. For the recommended composed instance the corresponding residues are 36, 63, and 258712. Its independent bound-and-binding audit also reconstructs every weighted coefficient and checks the new actual fixed payloads.

The native backend is checked independently without importing the generator. The audit reconstructs all four integers and checks 3,772 nonzero edge codes, 59,902 native-basis identities, both rotation intersections, carry and padding bounds, and the prime-power controller conditions on the actual new coefficients. A separate mathematical review checks the order of the alignment and occupancy arguments.

Clean rebuilding and byte comparison establish that the recorded circuits are the generator outputs. They do not decide whether the equation has a solution. The modular random assignments are transcription diagnostics; the coefficient reconstruction, norm identities, finite injectivity enumeration, and leading-coefficient certificate have exact stated meanings.

6.1 External and inherited dependencies

The proof uses the pinned K215 and compiler equivalences, Sun's 2021 Lucas and binomial results, Sun's 2026 ten-integer bridge and interpolation theorem, and elementary finite-field, Pell, and unique-factorization facts. The 2026 paper remains a preprint dependency. The new code and adapter have not been kernel-checked end to end in Lean or Rocq.

The comparison ten-integer polynomial remains bound to the original complete tuple. The recommended variant in Section 5 is fully rebound: all 402 weighted coefficient recipes, their bounds, the actual source-base payload, and the final degree certificate are regenerated from the new tuple. Separate independent audits check both instances. The smaller tuple is also concretely bound to P67 and the retained system.

6.2 What the conclusions do and do not say

The theorems are provable equivalences with consistency, not proofs of consistency. The new degree reduction is established relative to the explicitly pinned predecessor, and the coefficient reduction is established relative to the complete package-36 tuple. Neither is a claim that no smaller equation exists. Classical Bose–Chowla sets and Sun's universal theorem retain their attribution. The synthesis and explicit instance improvements are the new contributions presented here; a limited source search is not evidence of worldwide priority.

7 Further research questions

Can the actual source support use a shorter additive code? A full B_4 code separates all 52,360 quartic monomials, while the source has only 402 nonzero records. It is enough that each supported exponent avoid every competing degree-at-most-four monomial. An exact support-aware search could lower the maximum position substantially. Checking collisions only between two supported monomials would be insufficient: an unsupported monomial from the fourth power could still contaminate the coefficient being recovered.

Can span and circuit cost be optimized jointly? The former powers-of-five positions are unusually cheap to share. The new irregular positions achieve a much shorter span but increase the gate count. Codeword translation, assignment to parent variables, and shared addition chains can be searched together. A useful comparison must state both span and emitted operation count.

Can the Jones nine-natural-variable route use additive coding? The related coefficient-extraction stage makes this plausible, but each bound, mask, and reverse implication needs a separate audit. The ten-integer proof does not automatically certify a nine-variable circuit after a textual substitution. This is the most direct route to combining the new degree improvement with the collection's smallest natural-variable count.

Can other native backends benefit from the same adapter? Section 5 completes the composition of the two present improvements. The same architecture permits future coherent compiler tuples, but each replacement must regenerate its source coefficient bounds and fixed leaves. Automating that interface, with proof certificates rather than only construction scripts, would make subsequent backend comparisons much easier to audit.

Can the input quartic use fewer positive variables? The identity $D^2 + c^2 = (Ac)^2 + 1$ suggests removing a Pell discriminant, but in the present system it merely moves the auxiliary cost. Eliminating the index quotient runs into divisibility and proof-order obstacles involving $\gcd(E, H)$. In particular $X = 2^R$ cannot be used before the argument establishing the packed index has recovered it. No reduced quartic count or eleven-positive-variable construction is asserted here.

Which formal lemma should come first? The most reusable addition to the formal development is the additive coefficient-extraction lemma parameterized by a finite injectivity certificate, followed by the mask and source interface. This separates the new proof from the existing Lucas and machine endpoints. An exact circuit checker can verify degree upper bounds and the nonzero modular leading coefficient.

How close is the displayed code to a minimum? For an integer B_4 code with 32 positions in $[0, T]$, counting gives $4T + 1 \geq \binom{35}{4}$, hence $T \geq 13090$. This lower bound is weak and does not show that 942302 is close to optimal. Optimizing this finite code is a concrete combinatorial task. Proving a globally simplest Diophantine consistency equation is a separate problem.

Can occupancy information be compressed further? The new per-edge widths are minimal under the stated separate injective nonzero-word interface. This leaves open joint encodings, different tile presentations, alternative support geometry, and larger changes to the computation-to-polynomial compiler. A claimed gain must preserve both the local language and the global alignment/halting theorem.

8 Reproduction and incorporation

The top-level `reproduce.py` runs the construction and finite audits using Python's standard library. The report is built with `latexmk-pdfreport.tex`. The directory includes:

- the complete LaTeX article and its compiled PDF;
- the composed ten-integer circuit, the original-tuple comparison instance, and four alternative finalizers for that comparison;
- the literal parent, coefficient recipes, and compressed fixed integers;
- the new native backend, its concrete P67/system bindings, and independent coefficient audit;
- finite verification receipts, independent mathematical reviews, source pins, notices, and licenses.

The package is intended as a new sibling in the user's `con-zfc-diophantine-forms` collection. Existing source numbers and old tuples should retain their identities. The supplied fixed code positions describe a new polynomial; they are not an edit to the old package-31 polynomial's degree.

A The literal positive parent system

Use the variables and affine abbreviations of Section 2, and write $R = jc - V$ as a computed expression. The 22 equations below specify the parent of the two ten-integer instances. Their residuals are left side minus right side.

$$\begin{aligned} L &= q^2 - 1, & N &= q(L+1), \\ X &= wq, & a &= E + sN, \\ E &= X(a - E), & c &= k(a - E) + \eta, \\ P &= E(c - \eta) + k, & \tau^2 &= P(P - k) + 1, \\ D &= X + ac + \gamma H, & \Delta &= a^2 + H, \\ C_2 &= c^2, & D^2 &= \Delta C_2 + 1, \\ k &= R + 1 + hE. \end{aligned}$$

The next two transport equations are

$$\begin{aligned} R &= (q - F)(N - q) - CL + (M_C + qM_F)J, \\ (K_0 + w)C + (q - F) - z(q - 1) &= 1. \end{aligned}$$

Finally,

$$\begin{aligned} F_2 &= f^2, & V &= of - c, \\ g &= y^2 - V^2, & I &= iC_2, \\ B_{\text{aux}} &= I^2, & B_{\text{aux}} &= \Delta(F_2 - 1), \\ y^2 - B_{\text{aux}} g &= 1. \end{aligned}$$

The transport coordinate here is the original unshifted one. The later 60-operation quadratic system uses a shifted transport coordinate; its residual list must not be silently substituted when reconstructing this package's coefficient records. The JSON source is authoritative and is checked against the displayed equations.

References

[1] V. Reshetnikov, *ProveIt*, revision `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`, 23 October 2026. Collection: https://github.com/VladimirReshetnikov/ProveIt/tree/22e75dd6b87b1b199bb08c778d7cf075ae5aae08/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms. The source provenance ledger identifies individual paths and blobs.

[2] ProveIt package `31-sun10`, *Smaller explicit Diophantine forms for Con(ZFC)*, at revision [1], particularly `signed10/SIGNED10_PROOF.md`, `coefficients/COEFFICIENT_PROOF.md`, and `signed10/data/parent_quartic31.json`. Its ten-integer finalizer improvements are inherited here.

[3] ProveIt package `36-small-tuple`, *Smaller Diophantine presentations of ZFC consistency*, at revision [1]. The native compiler, actual-rank padding method, and 59-operation system are the baseline for the separate coefficient improvement.

[4] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics **64** (2021), 281–306. https://arxiv.org/abs/1704.03504v7. The source coding and Lucas/binomial results are used with explicit hypotheses.

[5] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594v2. Proposition 2.1, Theorems 3.1 and 4.1, Propositions 6.1–6.2, and Section 7. This is a preprint dependency.

[6] R. C. Bose and S. Chowla, *Theorems in the additive theory of numbers*, Commentarii Mathematici Helvetici **37** (1962/63), 141–147. https://doi.org/10.1007/BF02566968. Earlier author report: https://www.cs.umd.edu/~gasarch/COURSES/858/S13/BoseChowla.pdf.

[7] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026; first version 2025. https://arxiv.org/abs/2506.20909v3. Equations (1.6)–(1.7) and Definition 2.1 give an independent exposition of the positional source architecture.